Sinkhorn revisited - joint calibration to European and One-Touch prices

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Abstract

We apply a Sinkhorn scheme to the problem of constructing an admissible joint law for X_T and the supremum S_T of a continuous martingale X, given target laws μ, ν for X_T, S_T. This corresponds to fitting European and One-Touch prices at all strikes and barrier levels at maturity T. ν necessarily lies between the classical Azéma-Yor and Perkins bounds, but we show this condition is not sufficient with a simple counterexample. After each Sinkhorn cycle, the law of S_T and the Rogers[Rog93] condition are satisfied exactly, and the X_T-marginal converges to μ (under a finite-entropy feasibility assumption) at rate $1/\sqrt{n}$ in the total variation metric. This can also be solved as an optimal transport linprog problem as in [DR15] where one computes the max or min price of a non-tradable option given tradable options, but the latter typically leads to a sparse joint law which may be less realistic from a financial standpoint. We show that a (transformed) left-curtain coupling gives an explicit solution to the problem under additional restrictions on ν, and compute a sharp upper price bound for barrier options with supermodular payoffs when the continuity assumption on X is dropped.

1 Background

In the seminal work of [Rog93], the following fundamental result is established [1]:

Theorem 1.1 *(cf. Theorem 3.1 and Lemma 2.3 in [Rog93]; see also Theorem 3.1 in [DR15]). Fix a $T > 0$ and $x_0 \in \mathbb{R}$. A probability measure π on $\{(s,x) : -\infty < x \leq s,\ s \geq x_0\}$ is the joint law of (S_T, X_T) (where $S_t = \sup_{0\leq u\leq t} X_u$) for some continuous martingale X with $X_0 = x_0$ if and only if $\int |x|\pi(ds,dx) < \infty$, $\int x\pi(ds,dx) = x_0$, and*

$$\eta(ds) = s\nu(ds) - \bar{F}_\nu(s)ds \qquad \text{(as measures on } (x_0,\infty)), \tag{1}$$

where $\nu(ds) = \int_{x\leq s} \pi(ds,dx)$, $\eta(ds) = \int_{x\leq s} x\pi(ds,dx)$, and $\bar{F}_\nu(s) = \nu((s,\infty))$.

Eq (1) has the physical interpretation:

$$\mathbb{P}(S_T > b)db = \mathbb{E}((S_T - X_T)1_{S_T\in db}) = \mathbb{E}((b - X_T)1_{S_T\in db}) \tag{2}$$

and we give a sketch proof of the sufficiency part of the [Rog93] proof in Appendix D which uses basic excursion theory.

In Eq 3.8 in [DR15], the characterisation in Theorem 1.1 is re-stated in the more intuitive integrated form: for π supported on $\{x \leq s,\ s \geq x_0\}$ with $\int |x|\pi(ds,dx) < \infty$,

$$\int_{\{s\geq a\}} (x-a)\pi(ds,dx) = 0 \qquad \text{for all } a \geq x_0, \tag{3}$$

i.e. $\mathbb{E}(X_T - a;\ S_T \geq a) = 0$ for all $a \geq x_0$, which is clearly a necessary condition from the optional stopping theorem, and the tail condition $\lim_{a\to\infty} a\bar{F}_\nu(a) = 0$ follows from dominated convergence (see just below Eq 3.8 in [DR15]). Differentiating (3) (in the sense of measures) with respect to a leads to Eq (2).

Example 1.1 (Brownian motion stopped at a lower barrier) *(see Remark (ii) after Theorem 3.1 of [Rog93])*[2]*. Let W be a standard Brownian motion with $W_0 = 0$, let $\tau = \inf\{t \geq 0 : W_t = -1\}$, and let $X_t = W_{t\wedge\tau}$. Then X is a continuous local martingale with $X_\infty = -1$ almost surely, and its supremum $S_\infty = \sup_{t\geq 0} X_t$ has law $\nu(ds) = \frac{ds}{(1+s)^2}$ so $\bar{F}_\nu(s) = \nu((s,\infty)) = \frac{1}{1+s}$. The joint law of (S_∞, X_∞) satisfies the local Rogers condition (1) on $(0,\infty)$, but X is not uniformly integrable: $\mathbb{E}(X_\infty) = -1 \neq 0$ and $s\bar{F}_\nu(s) \to 1$ as $s \to \infty$.*

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[1] See also Remark (i) after Theorem 3.1 in [Rog93]. The results in [Rog93] are formulated for uniformly integrable (UI) martingales on $[0,\infty)$ and their terminal value M_∞. They apply here because a martingale X on $[0,T]$, extended by $X_t = X_T$ for $t \geq T$, is a UI martingale on $[0,\infty)$ with $X_\infty = X_T$ and $\sup_{t\geq 0} X_t = S_T$, and conversely a UI continuous martingale can be time-changed onto $[0,T]$ without altering its terminal value or its supremum. Theorem 1.1 is thus a restatement of Theorem 3.1 and Lemma 2.3 in [Rog93], together with Remark (i) there for $x_0 \neq 0$.

[2] See Theorem 4.1 in [Rog93] for another interesting related result

Proof. For $s \geq 0$, the event $\{S_\infty > s\}$ is the event that W reaches s before -1, which has probability $1/(1+s)$ (the gambler's ruin probability, obtained by applying the optional stopping theorem to W stopped at the first exit from $(-1, s)$); this gives $\bar{F}_\nu$ and, on differentiating, the density $\nu(s) = (1+s)^{-2}$. Since $X_\infty = -1$ on every path, the measure $\eta(ds) = \int x\pi(ds, dx)$ is $\eta(ds) = -\nu(ds)$. Hence, for $s > 0$,

$$s\nu(s) - \bar{F}_\nu(s) = \frac{s}{(1+s)^2} - \frac{1}{1+s} = \frac{s-(1+s)}{(1+s)^2} = -\frac{1}{(1+s)^2} = -\nu(s)$$

so $\eta(ds) = s\nu(ds) - \bar{F}_\nu(s)ds$ on $(0, \infty)$, which is Eq (1). On the other hand $\mathbb{E}X_\infty = -1 \neq X_0$, so X is not a UI martingale, and correspondingly $s\bar{F}_\nu(s) = s/(1+s) \to 1 \neq 0$, so the tail condition of Theorem 1.1 fails. ■

2 Outline of problem

Our aim is to construct a continuous martingale X with prescribed marginals μ and ν for X_T and S_T. We refer to (1) as the local Rogers condition, and this condition will form the basis of the Sinkhorn scheme below.

Based on Theorem 1.1, we impose the following natural conditions:

Assumption 2.1 *$\int |x|\mu(dx) < \infty$, $\int x\mu(dx) = x_0$ and $\lim_{a\to\infty} a\bar{F}_\nu(a) = 0$, and ν lies between the Perkins and Azéma-Yor (AY) model-independent bounds, i.e.*

$$\sup_{K<B} \{\mu([B,\infty)) + \frac{C_\mu(B) - P_\mu(K)}{B-K}\} \;\leq\; \nu([B,\infty)) \;\leq\; \inf_{K<B} \frac{C_\mu(K)}{B-K} \tag{4}$$

[3] *for all $B \geq x_0$, where $C_\mu(K) = \int (x-K)^+\mu(dx)$ and $P_\mu(K) = \int (K-x)^+\mu(dx)$ denote the call and put pricing functions.*

Definition 2.1 *A probability measure on $\{(s,x) : -\infty < x \leq s < \infty, s \geq x_0\}$ is called **admissible** if it has X-marginal μ, S-marginal ν, and satisfies the conditions in Theorem 1.1.*

Lemma 2.2 *(Convex-order necessary condition). Suppose an admissible joint law π of (S_T, X_T) exists, and let $\eta(ds) = \int_{x\leq s} x\,\pi(ds, dx)$. Then:*
(i) $\eta \ll \nu$, so that $b := d\eta/d\nu$ is well defined ν-almost everywhere and $\mathbb{E}(X_T|S_T) = b(S_T)$; if ν has a density, $b(s) = s - \bar{F}_\nu(s)/\nu(s)$.
(ii) $\mu \succeq_{\text{cx}} \nu \circ b^{-1}$, that is, $\int \phi\, d\mu \;\geq\; \int \phi(b(s))\,\nu(ds)$ for all convex $\phi : \mathbb{R} \to \mathbb{R}$, where $\nu \circ b^{-1}$ is the law of $b(S_T)$.

Proof.
(i) For any Borel set A with $\nu(A) = 0$ we have $\pi(A \times \mathbb{R}) = \mathbb{P}(S_T \in A) = 0$, hence $\eta(A) = \mathbb{E}\big(X_T 1_{\{S_T \in A\}}\big) = 0$; this uses $\mathbb{E}|X_T| < \infty$. Thus $\eta \ll \nu$, and the Radon–Nikodym derivative $b = d\eta/d\nu$ exists. By definition of η, $\int_A b\, d\nu = \eta(A) = \mathbb{E}\big(X_T 1_{\{S_T \in A\}}\big)$ for every Borel A, which is the statement $\mathbb{E}(X_T \,|\, S_T) = b(S_T)$. When ν has a density, the local Rogers condition (1) gives $\eta(ds) = (s\nu(s) - \bar{F}_\nu(s))ds$ and hence the formula for b.

(ii) For convex ϕ, the tower property and the conditional Jensen inequality give

$$\int \phi\, d\mu = \mathbb{E}\,\phi(X_T) = \mathbb{E}\big[\mathbb{E}(\phi(X_T) \,|\, S_T)\big] \;\geq\; \mathbb{E}\,\phi\big(\mathbb{E}(X_T \,|\, S_T)\big) = \mathbb{E}\,\phi(b(S_T)) = \int \phi(b(s))\nu(ds).$$

■

Proposition 2.3 (Sufficiency when b is increasing) *Suppose Assumption 2.1 holds, that ν has a density which is positive and continuously differentiable on the interior of its support, and that $s \mapsto 1/\bar{F}_\nu(s)$ is strictly convex there. Then $b(s) = s - \bar{F}_\nu(s)/\nu(s)$ is strictly increasing there. Let $Z = b(S_T)$ and let ν_Z be its law.*

(i) If $\mu \succeq_{\text{cx}} \nu_Z$, then there exists a martingale coupling $m(dz, dx) = \nu_Z(dz)m_z(dx)$ of ν_Z and μ, i.e. $\int x m_z(dx) = z$ for ν_Z-almost every z. For any such m, the joint law

$$\pi(ds, dx) = \nu(ds)\, m_{b(s)}(dx)$$

of (S_T, X_T) — draw $S_T \sim \nu$ and then X_T from the kernel of m at the point $z = b(S_T)$ — has marginals ν and μ and satisfies the local Rogers condition (1); π is admissible if and only if m_z is supported on $(-\infty, b^{-1}(z)]$ for ν_Z-almost every z.

[3] the bounds are sharp here, and attained by the Perkins and Azéma-Yor Skorokhod embeddings respectively, see the original papers [AY79],[Per86] or [Obl04],[Cox04],[Hob11] for details. See also [BHR01] and [HOST16] for the two (and n)-maturity extensions of Azéma-Yor, or [BBBL26], [HLMPT26] for more recent developments on Schrödinger bridges and Bass/Föllmer martingales.

(ii) Suppose in addition that μ has a density, that the pair (ν_Z, μ) satisfies the regularity conditions of [HLT16] (Assumptions 3.3 and 3.6 there), and that $\nu_Z((-\infty, z_0)) = 0$, where $z_0 = b(x_0)$, and there exists $z_* > z_0$ such that $f_{\nu_Z} > f_\mu$ on (z_0, z_*) and $f_{\nu_Z} < f_\mu$ on (z_*, r_{ν_Z}), where r_{ν_Z} is the right endpoint of the support of ν_Z. This single-crossing condition ensures that the left-curtain coupling has no identity part, as in Example 2.1. If the left-curtain coupling, whose branches $x_-(s) \le x_+(s)$ are determined by

$$\int_{x_-(s)}^{x_+(s)} \mu(dx) = 1 - \bar{F}_\nu(s), \qquad \int_{x_-(s)}^{x_+(s)} x\mu(dx) = \int_{x_0}^{s} b(u)\nu(u)du,$$

satisfies $x_+(s) \le s$ for ν-almost every s, then (μ, ν) is admissible, an admissible law being

$$\pi(dx|s) = q(s)\delta_{x_+(s)} + (1 - q(s))\,\delta_{x_-(s)}, \qquad q(s) = \frac{b(s) - x_-(s)}{x_+(s) - x_-(s)}.$$

Proof. See Appendix E. ■

Example 2.1 (An explicit family) *Let $\mu \sim U(-1, 1)$, so $x_0 = 0$, and let ν have tail $\bar{F}_\nu(s) = (1 - s)(1 - \theta s)$ on $[0, 1]$ with $0 < \theta \le \frac{1}{2}$, so that $\nu(s) = 1 + \theta - 2\theta s$. Then $\bar{F}_\nu$ lies between the Azéma-Yor bound $1 - s$ and the Perkins bound $1 - \sqrt{s}$, and touches the latter only in its slope at $s = 1$ when $\theta = \frac{1}{2}$. Since $\log(1/\bar{F}_\nu) = -\log(1-s) - \log(1-\theta s)$ is convex, $1/\bar{F}_\nu$ is strictly convex and, by Proposition 2.3, $b(s) = s - \frac{(1-s)(1-\theta s)}{1+\theta-2\theta s}$ is strictly increasing, from $b(0) = -1/(1 + \theta)$ to $b(1) = 1$. All the quantities in Proposition 2.3 are then explicit. Since μ has density $\frac{1}{2}$, the two equations for the branches read $\frac{1}{2}(x_+ - x_-) = 1 - \bar{F}_\nu(s)$ and $\frac{1}{4}(x_+^2 - x_-^2) = \int_0^s b(u)\nu(u)du$.*

The branches are monotone, x_- decreasing from $b(0)$ to -1 and x_+ increasing from $b(0)$ to 1, and the support condition $x_+(s) \le s$ reduces to $2\theta s + \theta^2 s(1 - s) \le 1$, whose left-hand side is increasing in s and equals 2θ at $s = 1$: it holds on all of $[0, 1]$ precisely when $\theta \le \frac{1}{2}$. Hence for every $\theta \in (0, \frac{1}{2}]$ the two-point law $\pi(dx|s) = q(s)\delta_{x_+(s)} + (1 - q(s))\delta_{x_-(s)}$ is an explicit admissible joint law of (S_T, X_T), and the pair (μ, ν) is admissible. The threshold $\theta = \frac{1}{2}$ is the same as the one at which $\bar{F}_\nu$ ceases to dominate the Perkins bound at $s = 1$.

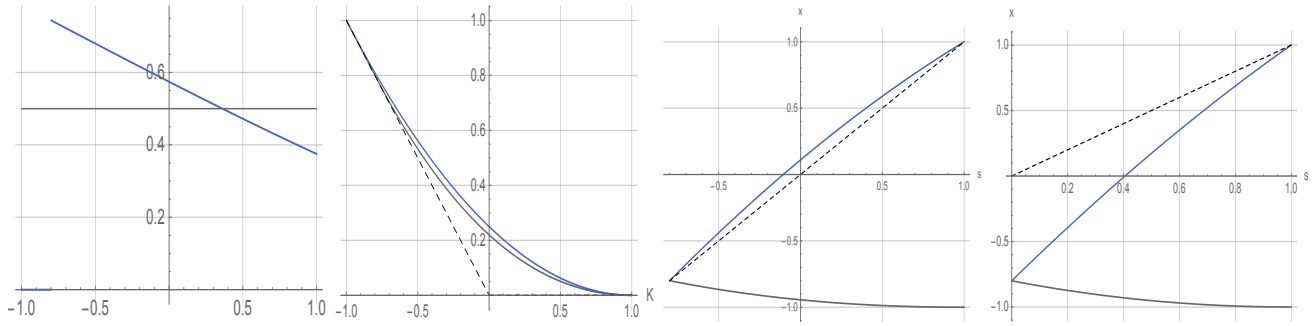


Figure 1: In the 1st plot we plot the densities of Z and X and in the 2nd plot we check the convex order condition for Example 2.1 by pricing call options on Z for a range of strike values K (solid grey) and X (blue). In the 3rd plot we see the upper and lower branches $T_u(z)$ (blue) and $T_d(z)$ (solid grey) for the left-curtain coupling, and on the right we see the corresponding $x_+(s)$ (blue) and $x_-(s)$ (solid grey), and we (correctly) see that $x_+(s) \le s$.

As alluded to in the abstract, the following counterexample shows that ν being a probability measure and lying between the Perkins and Azéma-Yor bounds is not a sufficient condition for an admissible joint law to exist.

Counterexample 2.1 *Let $\mu \sim U(-1, 1)$, so $x_0 = 0$, and let ν be the law on $(0, 1)$ with tail cdf*

$$\bar{F}_\nu(s) = 1 - s - \tfrac{7}{2}\, s(1 - s)^5, \qquad 0 \le s \le 1.$$

The Azéma–Yor and Perkins bounds for μ are $1 - s$ and $1 - \sqrt{s}$ respectively, and $1 - \sqrt{s} \le \bar{F}_\nu(s) \le 1 - s$ on $(0, 1)$, so ν satisfies Assumption 2.1, but there is no continuous martingale X with $X_T \sim \mu$ and $S_T \sim \nu$.

Proof. Suppose such a martingale exists, and let π be the joint law of (S_T, X_T). By Theorem 1.1, π satisfies (1): $\eta(ds) = s\nu(ds) - \bar{F}_\nu(s)\, ds$ on $(0, \infty)$, where $\eta(ds) = \int_{x \le s} x\, \pi(ds, dx)$. Since $X_T \ge -1$ almost surely, $\eta(ds) \ge -\int_{-1 \le x \le s} \pi(ds, dx) = -\nu(ds)$. Substituting the expression for η from Eq (1) into the left-hand side, this reads $s\nu(ds) - \bar{F}_\nu(s)\, ds \ge -\nu(ds)$, i.e.

$$(1 + s)\, \nu(ds) \;\ge\; \bar{F}_\nu(s)\, ds \qquad \text{on } (0, 1),$$

so $(1 + s)\nu(s) \ge \bar{F}_\nu(s)$ for Lebesgue almost every $s \in (0, 1)$. But this inequality is violated for $\nu(s) = -\bar{F}'_\nu(s) = 1 + \frac{7}{2}(1 - s)^4(1 - 6s)$ on a neighbourhood of $s = \frac{1}{3}$. ■

Motivated by financial applications, we henceforth assume μ only has support on the positive real line. If X is an asset price and interest rates and dividends are zero, μ can be inferred from European option prices at all

strikes via the Breeden-Litzenberger formula (using e.g. an SVI smile interpolation in practice as in Figure 7), and $\nu([B,\infty))$ is the price of a One-Touch option on X with barrier level $B > X_0$. Barrier options are most actively traded in FX options and are predominantly over-the-counter instruments.

2.1 Sinkhorn integral equations

From here on, we further assume that μ, ν have densities with respect to the Lebesgue measure, and set

$$r(s) = s\nu(s) - \bar{F}_\nu(s)$$

i.e. the density on the right-hand side of (1).

Let $p(s,x)$ be a candidate joint density for (S,X). Based on the $\mathbb{E}(X_T 1_{S_T \in db})$ term in (2) and the entropic optimal transport literature, we make the exponential ansatz:

$$p(s,x) = R(s,x)e^{u(x)+v(s)+x\lambda(s)}$$

where R is a reference density (see Assumption 2.4 for conditions on R). Then the X-marginal condition is

$$\mu(x) = e^{u(x)} \int_x^\infty R(s,x)e^{v(s)+\lambda(s)x} ds$$

which we can rearrange as

$$u(x) = \log \mu(x) - \log(\int_x^\infty R(s,x)e^{v(s)+\lambda(s)x} ds) \tag{5}$$

if the integral term lies in $(0,\infty)$ (this will be addressed in Lemma C.1). Similarly, the S-marginal condition is

$$\nu(s) = e^{v(s)} \int_0^s R(s,x)e^{u(x)+\lambda(s)x} dx$$

and the local Rogers condition in (1) is

$$r(s) = e^{v(s)} \int_0^s xR(s,x)e^{u(x)+\lambda(s)x} dx.$$

Dividing by the S-marginal equation then gives

$$\frac{\int_0^s xR(s,x)e^{u(x)+\lambda(s)x} dx}{\int_0^s R(s,x)e^{u(x)+\lambda(s)x} dx} = b(s) = \frac{r(s)}{\nu(s)} \tag{6}$$

where $b(s)$ is defined for ν-almost every s. Under suitable technical conditions (see Lemma 2.5 for details), $\lambda(s)$ can be computed from (6) as a one-dimensional root-finding exercise. The S-marginal equation gives

$$v(s) = \log \nu(s) - \log(\int_0^s R(s,x)e^{u(x)+\lambda(s)x} dx). \tag{7}$$

Hence we see that (5), (6) and (7) are the three coupled equations for u, λ and v. See [Guy20] and [For26] for a similar approach to the SPX-VIX and FX cross-smile calibration problems respectively.

2.2 The Sinkhorn scheme

The coupled equations above suggest the following iterative scheme. Set $u_0 = v_0 = \lambda_0 = 0$. Given (u_n, v_n, λ_n), first set

$$u_{n+1}(x) = \log \mu(x) - \log(\int_x^\infty R(s,x)e^{v_n(s)+\lambda_n(s)x} ds). \tag{8}$$

This enforces the X-marginal exactly. Next, choose $\lambda_{n+1}(s)$ as the solution of

$$\frac{\int_0^s xR(s,x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} dx}{\int_0^s R(s,x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} dx} = b(s). \tag{9}$$

This is a scalar root-finding problem in λ for each s; existence and uniqueness of the root for ν-almost every s are established in Lemma 2.5. Finally set

$$v_{n+1}(s) = \log \nu(s) - \log(\int_0^s R(s,x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} dx). \tag{10}$$

The next u-update restores the X-marginal, which is the only constraint disturbed by the completed (λ, v) update. In practice, one iteration therefore consists of an explicit u-update, one scalar root-find for $\lambda_{n+1}(s)$ for ν-almost every s, and an explicit v-update.

For probability measures P and Q we write $H(P|Q) = \int \log \frac{dP}{dQ}\, dP$ if $P \ll Q$ and $H(P|Q) = +\infty$ otherwise. The integral is well defined in $[0, \infty]$: writing $\rho = dP/dQ$, $\int (\log \rho)^- \, dP = \int \rho (\log \rho)^- \, dQ \le 1/e$, since $\rho \log \rho \ge -1/e$ for $\rho \ge 0$, and $H(P|Q) \ge 0$ by Jensen's inequality. [4]

From here on we make the following further assumption:

Assumption 2.4 *$\mu = 0$ outside $0 \le x \le K$, $\nu = 0$ outside $X_0 \le s \le K$ for some $K < \infty$ [5], and $\mu(x) > 0$ for Lebesgue almost every $x \in (0, K)$ and $\nu(s) > 0$ for Lebesgue almost every $s \in (X_0, K)$. The reference density R is a probability density supported on*

$$s \ge X_0, \qquad 0 < x \le s$$

with $R(s,x) > 0$ for every $s \ge X_0$ and Lebesgue almost every $x \in (0, s)$. Finally, there exists an admissible probability density p^* such that $H(p^*|R) < \infty$.

Since $H(p^*|R) < \infty$, we have $p^* \ll R$, so $0 < X \le S$ p^*-almost surely. Since p^* is a density, $p^*(X = S) = 0$. Hence, using $b(s) = \mathbb{E}_{p^*}(X|S = s)$, (by the local Rogers condition), we obtain

$$0 < b(s) < s \tag{11}$$

for ν-almost every s.

Remark 2.1 (Reference density for Geometric Brownian motion) A natural choice for the reference density R is the joint law of (S_T, X_T) for driftless geometric Brownian motion $X_t = X_0 e^{\sigma W_t - \sigma^2 t/2}$. $Y_t = \log X_t$ is a Brownian motion with drift $-\sigma^2/2$ and volatility σ started at $\log X_0$, and the reflection principle and Girsanov's theorem give the joint density $f_{M_T, Y_T}(b, y)$ of $M_T = \max_{t \le T} Y_t$ and Y_T. Setting $s = e^b$, $x = e^y$ we find that

$$R(s,x) = \frac{2\,\ell(s,x)}{\sigma^3 \sqrt{2\pi T^3}\; s\, x} \sqrt{\frac{X_0}{x}}\; e^{-\frac{\ell(s,x)^2}{2\sigma^2 T} - \frac{\sigma^2 T}{8}}, \qquad \ell(s,x) = \log \frac{s^2}{x X_0}, \tag{12}$$

for $x > 0$ and $s \ge \max(x, X_0)$, and $R(s,x) = 0$ otherwise. Its S-marginal is

$$\bar{F}_R(B) = \int_B^\infty \int_0^s R(s,x)\, dx\, ds = \Phi\Big(\frac{-\log(B/X_0) - \sigma^2 T/2}{\sigma\sqrt{T}}\Big) + \frac{X_0}{B}\, \Phi\Big(\frac{-\log(B/X_0) + \sigma^2 T/2}{\sigma\sqrt{T}}\Big), \qquad B \ge X_0,$$

where Φ is the standard Normal distribution function. Since geometric Brownian motion is a continuous martingale, R satisfies the local Rogers condition (1) exactly, and it is positive on the interior of its support, as required in Assumption 2.4. Note however that R is not supported on $s \le K$, so $q_0 = R$ charges the region $\{x > K\}$; this is why several statements in Section 2 and Appendix C are restricted to $n \ge 1$.

Lemma 2.5 *Suppose Assumption 2.4 holds. Let u be measurable and finite Lebesgue almost everywhere on $(0, K)$ and suppose that, for ν-almost every s, there exists at least one finite value of λ for which $0 < \int_0^s R(s,x) e^{u(x) + \lambda x} dx < \infty$. Then for ν-almost every s there exists a unique finite $\lambda(s)$ such that*

$$\frac{\int_0^s x R(s,x) e^{u(x) + \lambda(s) x} dx}{\int_0^s R(s,x) e^{u(x) + \lambda(s) x} dx} = b(s). \tag{13}$$

Moreover, λ admits a measurable version, and for ν-almost every s, the integral $\int_0^s R(s,x) e^{u(x) + \lambda x} dx$ is positive and finite for every finite λ, and in particular for $\lambda = \lambda(s)$.

Proof. See Appendix A. ■

3 Convergence for the Sinkhorn scheme

For the convergence argument, define $p_{i,j,k}(s,x) = R(s,x) e^{u_i(x) + v_j(s) + \lambda_k(s) x}$. Thus one complete cycle is $p_{n,n,n} \longrightarrow p_{n+1,n,n} \longrightarrow p_{n+1,n,n+1} \longrightarrow p_{n+1,n+1,n+1}$, corresponding to the u-update, the λ-update and the v-update respectively. By Eq (8), $(p_{n+1,n,n})^X = \mu$, so $p_{n+1,n,n}$ is a probability density. By Eq (10), $(p_{n+1,n+1,n+1})^S = \nu$, so every completed iterate is also a probability density. Moreover, Eq (9) gives

$$\frac{\int_0^s x p_{n+1,n+1,n+1}(s,x) dx}{\int_0^s p_{n+1,n+1,n+1}(s,x) dx} = b(s).$$

Since the denominator is $\nu(s)$ by construction, we see that $\int_0^s x p_{n+1,n+1,n+1}(s,x) dx = b(s)\nu(s) = r(s)$. Hence every completed iterate $p_{n,n,n}$ with $n \ge 1$ not only has the correct S-marginal but satisfies the local Rogers condition. The intermediate density $p_{n+1,n,n+1}$ need not have mass one, so in the entropy calculation below the λ- and v-updates are treated together.

[4] The same bound justifies evaluating $H(P|Q)$ as a double integral computed one variable at a time whenever P and Q are given by densities.

[5] note there is mild overloading of notation here; this K is not the strike price variable K in Assumption 2.1

Lemma 3.1 *For every $n \geq 0$ for which the u-update is well defined,*

$$\frac{p_{n+1,n,n}(s,x)}{p_{n,n,n}(s,x)} = \frac{\mu(x)}{(p_{n,n,n})^X(x)} \tag{14}$$

and

$$H(p_{n+1,n,n}|p_{n,n,n}) = H(\mu|(p_{n,n,n})^X). \tag{15}$$

Proof. Since $p_{n+1,n,n}$ and $p_{n,n,n}$ differ only in the u-potential, $p_{n+1,n,n}(s,x)/p_{n,n,n}(s,x) = e^{u_{n+1}(x)-u_n(x)}$. Integrating $p_{n,n,n}(s,x) = R(s,x)e^{u_n(x)+v_n(s)+\lambda_n(s)x}$ over s gives $(p_{n,n,n})^X(x) = e^{u_n(x)}\int_x^\infty R(s,x)e^{v_n(s)+\lambda_n(s)x}ds$, so Eq (8) can be written as $e^{u_{n+1}(x)} = \mu(x)e^{u_n(x)}/(p_{n,n,n})^X(x)$, which gives Eq (14). Hence, since $(p_{n+1,n,n})^X = \mu$,

$$H(p_{n+1,n,n}|p_{n,n,n}) = \int\int p_{n+1,n,n}(s,x)\log\frac{\mu(x)}{(p_{n,n,n})^X(x)}dxds = \int \mu(x)\log\frac{\mu(x)}{(p_{n,n,n})^X(x)}dx = H(\mu|(p_{n,n,n})^X),$$

which is Eq (15). ∎

Proposition 3.2 *Suppose Assumption 2.4 holds. Then $\|(p_{n,n,n})^X - \mu\|_{\mathrm{TV}} \longrightarrow 0$ as $n \to \infty$.* [6]

Proof. By Lemma C.1, every step of the scheme is well defined and $H(p^*|p_{n,n,n})$ and $H(p^*|p_{n+1,n,n})$ are finite for every $n \geq 0$. Adding Eqs (C-1) and (C-2), the term $H(p^*|p_{n+1,n,n})$ cancels and we obtain, for every $n \geq 0$,

$$H(p_{n+1,n,n}|p_{n,n,n}) + H(p_{n+1,n+1,n+1}|p_{n+1,n,n}) = H(p^*|p_{n,n,n}) - H(p^*|p_{n+1,n+1,n+1}). \tag{16}$$

Now sum Eq (16) over $n = 0, 1, \ldots, N-1$. On the right-hand side the sum telescopes, since the second term for index n cancels the first term for index $n+1$, leaving only the first term for $n = 0$ and the second term for $n = N-1$:

$$\begin{aligned}\sum_{n=0}^{N-1}(H(p_{n+1,n,n}|p_{n,n,n}) + H(p_{n+1,n+1,n+1}|p_{n+1,n,n})) &= \sum_{n=0}^{N-1}(H(p^*|p_{n,n,n}) - H(p^*|p_{n+1,n+1,n+1})) \\ &= H(p^*|p_{0,0,0}) - H(p^*|p_{N,N,N}) \\ &= H(p^*|R) - H(p^*|p_{N,N,N}) \leq H(p^*|R),\end{aligned}$$

since $p_{0,0,0} = R$ and entropy is non-negative. Letting $N \to \infty$ gives

$$\sum_{n=0}^{\infty}\big(H(p_{n+1,n,n}|p_{n,n,n}) + H(p_{n+1,n+1,n+1}|p_{n+1,n,n})\big) \leq H(p^*|R) \tag{17}$$

so in particular, $H(p_{n+1,n,n}|p_{n,n,n}) \longrightarrow 0$. By Lemma 3.1, $H(p_{n+1,n,n}|p_{n,n,n}) = H(\mu|(p_{n,n,n})^X)$. Therefore $H(\mu|(p_{n,n,n})^X) \longrightarrow 0$ and Pinsker's inequality gives $\|(p_{n,n,n})^X - \mu\|_{\mathrm{TV}} \longrightarrow 0$. ∎

3.1 Rate of convergence

Proposition 3.3 *Under the assumptions of Proposition 3.2, for every $n \geq 1$, $H((p_{n,n,n})^X|\mu) \leq \frac{H(p^*|R)}{n}$. Hence $\|(p_{n,n,n})^X - \mu\|_{L^1} \leq \sqrt{\frac{2H(p^*|R)}{n}}$, and $(p_{n,n,n})^X \to \mu$ in total variation at rate $O(n^{-1/2})$.*

Proof. Write $q_n = p_{n,n,n}$. Since $(p_{n+1,n,n})^X = \mu$, the Data Processing inequality (DPI, cf. Lemma 1.6 in [Nutz22]) gives

$$H(q_{n+1}^X|\mu) \leq H(q_{n+1}|p_{n+1,n,n}). \tag{18}$$

From Eq (14) $p_{n+1,n,n}(s,x)/q_n(s,x) = \mu(x)/q_n^X(x)$ for $n \geq 1$, so $q_n(s,x)/p_{n+1,n,n}(s,x) = q_n^X(x)/\mu(x)$ q_n-almost everywhere, since for $n \geq 1$ the density q_n^X is supported in $(0,K)$, where $\mu > 0$ almost everywhere. Hence

$$H(q_n|p_{n+1,n,n}) = \int\log\frac{q_n(s,x)}{p_{n+1,n,n}(s,x)}q_n(s,x)dsdx = \int\log\frac{q_n^X(x)}{\mu(x)}q_n(s,x)dsdx = \int\log\frac{q_n^X(x)}{\mu(x)}q_n^X(x)dx = H(q_n^X|\mu). \tag{19}$$

In particular, by Eq (18) with $n-1$ in place of n and Eq (17), $H(q_n|p_{n+1,n,n}) = H(q_n^X|\mu) \leq H(q_n|p_{n,n-1,n-1}) \leq H(p^*|R) < \infty$ for $n \geq 1$.

For $n \geq 1$, both q_n and q_{n+1} have S-marginal ν and conditional mean $b(s)$ by Lemma C.1. Moreover, as shown in the proof of Lemma C.1, $q = p_{n+1,n,n}$ satisfies the hypotheses of Lemma B.1, with $\lambda(s) = \lambda_{n+1}(s) - \lambda_n(s)$ and with $q' = p_{n+1,n+1,n+1} = q_{n+1}$. Hence Lemma B.1, with $p = q_n$, gives

$$H(q_n|p_{n+1,n,n}) = H(q_n|q_{n+1}) + H(q_{n+1}|p_{n+1,n,n}). \tag{20}$$

[6] see section 1.1 in [Nutz22] for the definition of the total variation distance $\|\cdot\|_{\mathrm{TV}}$ here

By Eq (20), $H(q_{n+1}|p_{n+1,n,n}) \leq H(q_n|p_{n+1,n,n})$.. Combining this with Eqs (18) and (19) gives

$$H(q_{n+1}^X|\mu) \leq H(q_{n+1}|p_{n+1,n,n}) \leq H(q_n|p_{n+1,n,n}) = H(q_n^X|\mu).$$

On the other hand, Eqs (18) and (17) give

$$\sum_{n=1}^{\infty} H(q_n^X|\mu) \leq \sum_{n=0}^{\infty} H(q_{n+1}|p_{n+1,n,n}) \leq H(p^*|R).$$

Since $H(q_n^X|\mu)$ is non-increasing and non-negative, $nH(q_n^X|\mu) \leq \sum_{k=1}^{n} H(q_k^X|\mu) \leq H(p^*|R)$, which proves the first claim. Finally, Pinsker's inequality gives $\|q_n^X - \mu\|_{L^1} \leq \sqrt{2H(q_n^X|\mu)} \leq \sqrt{\frac{2H(p^*|R)}{n}}$. ■

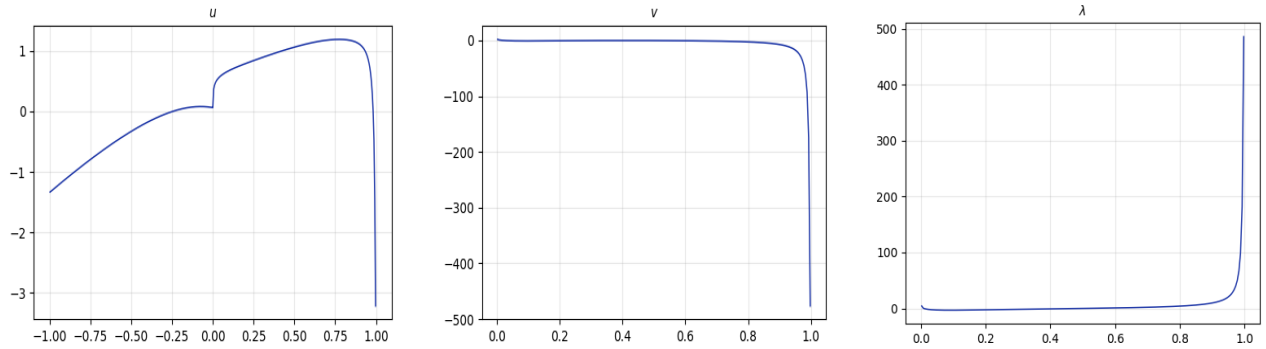


Figure 2: Here we have plotted u, v, λ for a target law ν which is a $(\frac{3}{5}, \frac{2}{5})$ weighting of the AY and Perkins laws respectively (so trivially an admissible (albeit singular) joint law exists) with $\mu \sim U[-1, 1]$. We take R to be the product measure $\mu \otimes \nu$, restricted to the region $x \leq s$ and renormalized, and we use 400 (equidistant) steps for the X direction and 200 for S.

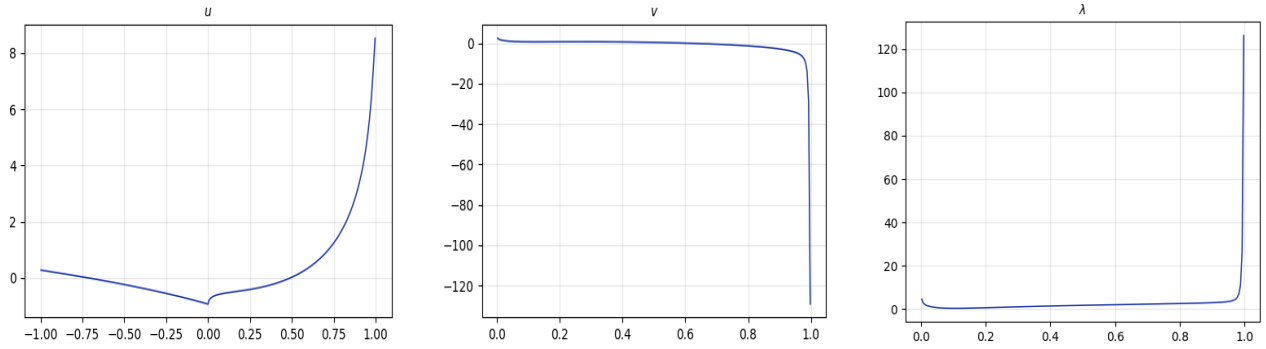


Figure 3: Here we have plotted u, v, λ for a target law ν with cdf $\nu([0, s]) = s^p$ for $s \in [0, 1]$ with $p = 3/4$, with all other parameters unchanged.

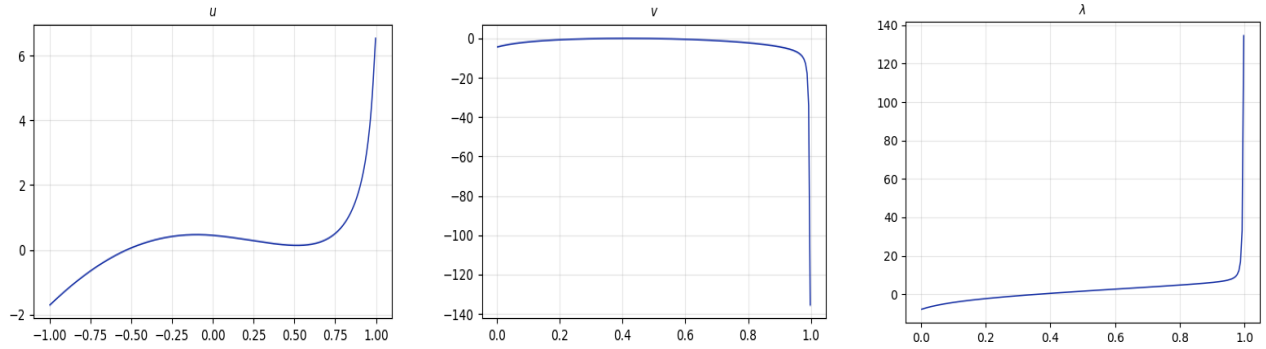


Figure 4: Here we have plotted u, v, λ for a target law ν with cdf $\nu([0, s]) = 1 - (1 - s)(1 - \theta s)$ for $s \in [0, 1]$ (see Example 2.1) with $\theta = 1/4$, with all other parameters unchanged.

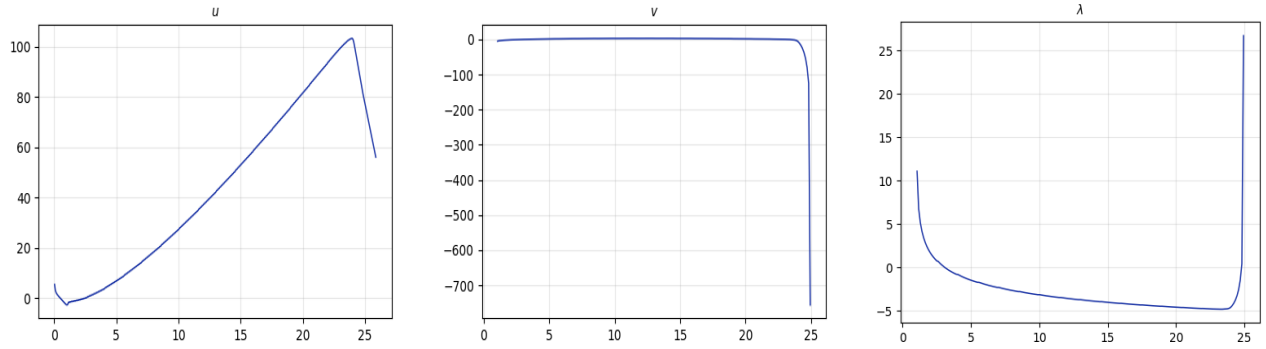


Figure 5: Here we have plotted u, v, λ for a target law ν which is a $(\frac{1}{3}, \frac{2}{3})$ weighting of AY and Perkins with $\mu \sim \text{Exp}(1)$ and R given by (12) with $X_0 = \sigma = T = 1$. In this case the AY optimal strike in (4) is $K^*(B) = B - 1$ and the Perkins optimal strike is $K_*(B) = -1 + B - W((Be^B - 1 - e^B)/e)$, where $W(.)$ denotes the principal branch of the Lambert-W function, i.e. the unique solution $w \geq -1$ of $we^w = z$ for $z \geq -1/e$

.

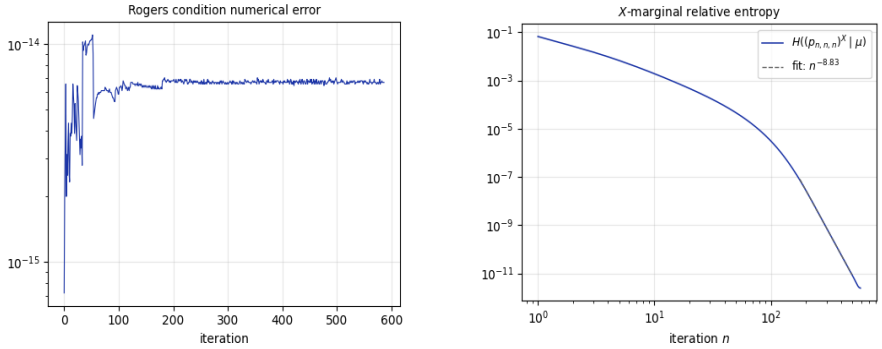


Figure 6: Here we have plotted the error for the Rogers condition for the final $\mu \sim \text{Exp}(1)$ example on the previous page (which is just discretization and root-finding error since theoretically this error is zero for the scheme by construction) and the (entropic) error for the X-marginal. The log-log slope over iterations 177-498 suggests that $H((p_{n,n,n})^X|\mu) \sim Cn^{-8.83}$, so the $O(\frac{1}{n})$ rate reported in Proposition 3.3 may not be sharp. The code takes about a minute to run on an old MacBook, but we ran to a lower tolerance than is typically necessary in practice.

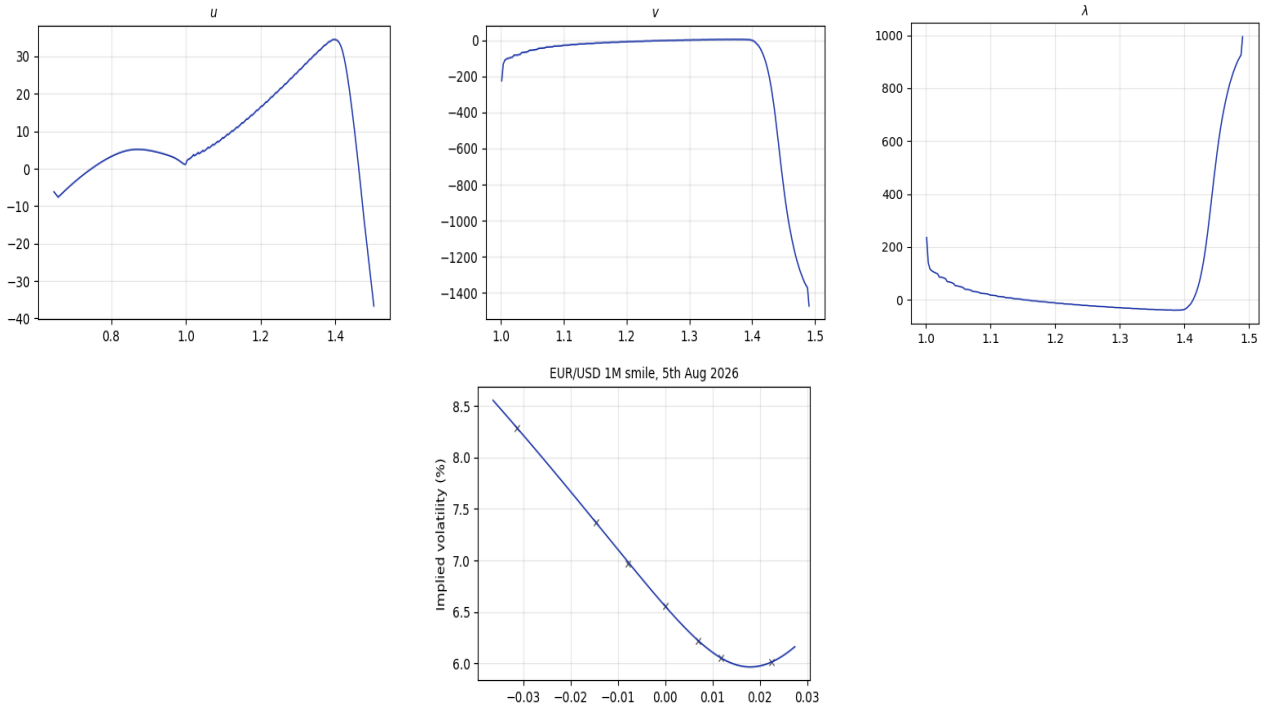


Figure 7: Here we have plotted u, v, λ with μ given by an SVI fit to the 1M EUR-USD smile here (the x-axis is log-moneyness here, see table below) with a (made-up) $(\frac{1}{3}, \frac{2}{3})$ AY-Perkins weighting as above and all other parameters unchanged from Figure 5. The parameters for the SVI fit are $(a, b, \sigma, \rho, m) = (0.00016844, 0.00805007, 0.01660353, 0.01493316, 0.01824409)$.

Strike	Call Price	Call Delta	Imp. Vol.
1.0888	0.03571	0.90	8.29%
1.1073	0.01966	0.75	7.37%
1.1148	0.01406	0.65	6.97%
1.1235	0.00852	0.50	6.56%
1.1314	0.00483	0.35	6.22%
1.1369	0.00299	0.25	6.05%
1.1490	0.00094	0.10	6.01%

Table 1: EUR/USD 1M smile on 5th Oct 2026, with spot EUR/USD at 1.1220, data taken from `https://www.investing.com/currencies/eur-usd-options`.

3.2 Non-continuous martingales and extremal prices for barrier options

If we relax the condition that X is continuous, the upper bound in (4) remains unchanged but the Perkins bound is replaced by the trivial lower bound $\mathbb{P}^{X\sim\mu}(\max(X, X_0) \geq B)$. In this case there is an explicit solution to the problem addressed in this paper described at the end of page 459 and page 460 in [Rog93] by setting $A_t \equiv \int_0^t 1_{\{B_u < c^{-1}(S_u)\}} du$, $\tau_t \equiv \inf\{u : A_u > t\}$ and $M_t = B_{\tau_t \wedge T}$ where $S_t = \sup_{0\leq u\leq t} B_u$ and $T = \inf\{u : B_u < b_\mu^{-1}(S_u)\}$ with $b_\mu(x) = \mathbb{E}^{X\sim\mu}(X|X \geq x)$ so T is the usual AY stopping time, and $M_\infty = B_T = b_\mu^{-1}(S_T)$ and $\bar{M}_\infty = c^{-1}(S_T)$ with

$c(s) = b_\mu(F_\mu^{-1}(F_\nu(s)))$. Here F_μ, F_ν denote the respective CDFs, and $x^+ \le c(x) \le b_\nu(x)$ so M_∞ and $\bar{M}_\infty$ are linked via the simple co-monotone coupling and X has positive-only jumps since c^{-1} is non-decreasing (see Fig 1 in [Rog93]). Note there is no need for a Sinkhorn scheme in this case unless one seeks a joint density for the terminal value and supremum.

Consider a payoff g such that $|g(s,x)| \le f(s) + h(x)$ for some $f \in L^1(\nu)$ and $h \in L^1(\mu)$, so $\mathbb{E}(|g(S,X)| < \infty)$ for every coupling of ν and μ. Then (by the rearrangement inequality) M maximises $\mathbb{E}(g(\bar{M}_\infty, M_\infty))$ over all martingales M with $M_\infty \sim \mu$ and $\bar{M}_\infty \sim \nu$ whenever g is *supermodular*, i.e. for every $x_1 < x_2$ the increment $s \mapsto g(s,x_2) - g(s,x_1)$ is non-decreasing in s; this holds for e.g. the up-and-in call payoff $g(s,x) = 1_{\{s \ge B\}}(x-K)^+$, whose increment is $1_{\{s\ge B\}}((x_2-K)^+ - (x_1-K)^+)$, zero for $s < B$ and a non-negative constant for $s \ge B$.

To see this in terms of a superhedge, note that for arbitrary s, x, if $a \ge K$,

$$1_{\{s\ge B\}}(x-K)^+ \le (x-a)^+ + (a-K)1_{\{s\ge B\}}.$$

Indeed, if $s < B$ the left-hand side is zero. If $s \ge B$, the inequality reduces to $(x-K)^+ \le (x-a)^+ + (a-K)$, which follows from $(u+v)^+ \le u^+ + v^+$ with $u = x-a$ and $v = a-K$.

For the co-monotone coupling this inequality becomes an equality. Assuming μ has a density as above, choose a so that $\mu((a,\infty)) = \nu((B,\infty))$. Then under the co-monotone coupling, $\{x \ge a\} = \{s \ge B\}$. Suppose first that $a \ge K$.

If $s < B$, then $x < a$, so $(x-a)^+ = 0$ and both sides are zero. If $s \ge B$, then $x \ge a \ge K$, so $(x-a)^+ + (a-K) = x - a + a - K = x - K = (x-K)^+$. Hence the superhedge is attained pathwise by the co-monotone coupling.

Finally, if $a < K$, then for arbitrary s, x, $1_{\{s\ge B\}}(x-K)^+ \le (x-K)^+$, so the European call itself is a superhedge. Under the co-monotone coupling, whenever $(x-K)^+ > 0$ we have $x > K > a$. Hence $x \ge a$ and $s \ge B$. Thus $1_{\{s\ge B\}}(x-K)^+ = (x-K)^+$, so the vanilla-call superhedge is again attained pathwise by the co-monotone coupling.

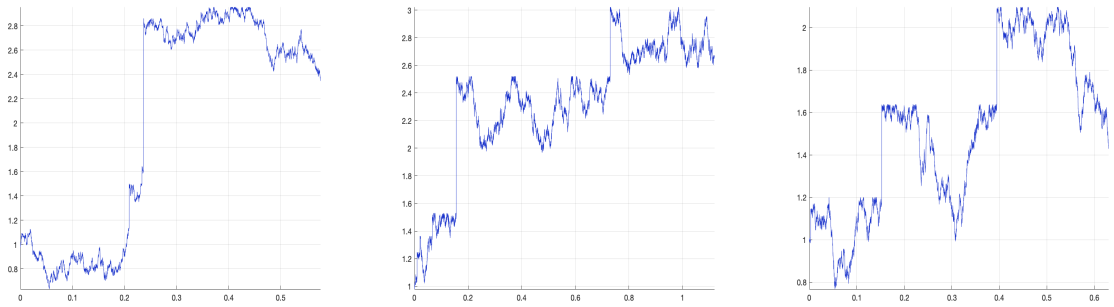


Figure 8: Here we see three simulations of the lost-time Rogers martingale M in Section 3.2 with target laws $\mu \sim \text{Exp}(1)$ and $\nu([b,\infty)) = \bar{F}_\theta(b) = \exp(-\frac{b(b-1)}{b-1+\theta})$ for $b \ge 1$ and $0 < \theta \le 1$ with $\theta = \frac{1}{2}$ here. At the endpoints, $\bar{F}_1(b) = e^{1-b}$ for $b \ge 1$, while $\bar{F}_0(1) = 1$ and $\bar{F}_0(b) = e^{-b}$ for $b > 1$, and the AY law here is $\text{Exp}(1) + 1$. Note for $\theta = 0$ we require an atom of mass $1 - e^{-1}$ at $b = 1$.

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A Proof of Lemma 2.5

Proof. Fix an s for which the hypothesis of the lemma holds, i.e. for which $Z_s(\lambda) \in (0,\infty)$ for some finite λ_2, where $Z_s(\lambda) = \int_0^s R(s,x)e^{u(x)+\lambda x}dx$; by assumption this is the case for ν-almost every s. Then, for any other finite λ_1,

$$Z_s(\lambda_1) = \int_0^s e^{(\lambda_1-\lambda_2)x} R(s,x)e^{u(x)+\lambda_2 x}dx.$$

Since $x \in (0,s)$, the factor $e^{(\lambda_1-\lambda_2)x}$ is bounded above and below by strictly positive constants. Hence (by comparison with the previous expression) $0 < Z_s(\lambda_1) < \infty$. Therefore $Z_s(\lambda)$ is positive and finite for every finite λ. Since $x \in (0,s)$, differentiation under the integral sign is justified locally in λ by dominated convergence. Moreover

$$\frac{d}{d\lambda}\log Z_s(\lambda) = \frac{\int_0^s xR(s,x)e^{u(x)+\lambda x}dx}{\int_0^s R(s,x)e^{u(x)+\lambda x}dx} =: m_s(\lambda),$$

which is the mean of x under the probability density $R(s,x)e^{u(x)+\lambda x}/Z_s(\lambda)$ on $(0,s)$, and (using the quotient rule) we can easily check that $m_s'(\lambda)$ is the corresponding variance,

$$m_s'(\lambda) = \int_0^s (x-m_s(\lambda))^2 \frac{R(s,x)e^{u(x)+\lambda x}}{Z_s(\lambda)}dx.$$

By Assumption 2.4, $R(s,x) > 0$ for Lebesgue almost every $x \in (0,s)$. Therefore, since $u(x)$ is finite Lebesgue almost everywhere, we have $\frac{R(s,x)e^{u(x)+\lambda x}}{Z_s(\lambda)} > 0$ for Lebesgue almost every $x \in (0,s)$. The integrand is strictly positive for Lebesgue almost every $x \in (0,s)$ except the single point $x = m_s(\lambda)$. Therefore

$$m_s'(\lambda) > 0$$

for every finite λ. Thus m_s is continuous and strictly increasing. Moreover, since $R(s,x)e^{u(x)} > 0$ for Lebesgue almost every $x \in (0,s)$, for every $\varepsilon \in (0,s)$ we have $\int_0^\varepsilon R(s,x)e^{u(x)}dx > 0$ and $\int_{s-\varepsilon}^s R(s,x)e^{u(x)}dx > 0$. Now fix $\lambda > 0$. Since

$$Z_s(\lambda) = \int_0^s R(s,x)e^{u(x)}e^{\lambda x}dx,$$

and the integrand is non-negative, $Z_s(\lambda) \geq \int_{s-\varepsilon}^s R(s,x)e^{u(x)}e^{\lambda x}dx$. For $x \in (s-\varepsilon, s)$, $e^{\lambda x} \geq e^{\lambda(s-\varepsilon)}$, so

$$Z_s(\lambda) \geq e^{\lambda(s-\varepsilon)} \int_{s-\varepsilon}^s R(s,x)e^{u(x)}dx.$$

Also, since $x < s$, $e^{\lambda x} \leq e^{\lambda s}$, and hence $Z_s(\lambda) \leq e^{\lambda s}\int_0^s R(s,x)e^{u(x)}dx = e^{\lambda s}Z_s(0)$. Therefore

$$e^{\lambda(s-\varepsilon)} \int_{s-\varepsilon}^s R(s,x)e^{u(x)}dx \leq Z_s(\lambda) \leq e^{\lambda s}Z_s(0).$$

Taking logarithms and dividing by λ gives

$$s - \varepsilon + o(1) \le \frac{\log Z_s(\lambda)}{\lambda} \le s + o(1)$$

as $\lambda \to \infty$. Since $\varepsilon > 0$ is arbitrary, $\lim_{\lambda\to\infty} \log Z_s(\lambda)/\lambda = s$. Similarly, using $\int_0^\varepsilon R(s,x)e^{u(x)}dx > 0$ and $e^{\lambda x} \ge e^{\lambda\varepsilon}$ for $x \in (0,\varepsilon)$ and $\lambda < 0$, $\lim_{\lambda\to-\infty} \frac{\log Z_s(\lambda)}{\lambda} = 0$. We now use the following elementary fact: for a differentiable convex function f on $\mathbb{R}$, the limits $\lim_{\lambda\to\pm\infty} f'(\lambda)$ exist in $[-\infty,\infty]$ and $\lim_{\lambda\to\pm\infty} f(\lambda)/\lambda = \lim_{\lambda\to\pm\infty} f'(\lambda)$. The function $f = \log Z_s$ is differentiable with $f' = m_s$, and convex because $f'' = m_s' \ge 0$ by the variance formula above. Applying this to $f = \log Z_s$ and combining it with the two limits just computed, we obtain

$$\lim_{\lambda\to-\infty} m_s(\lambda) = 0, \qquad \lim_{\lambda\to\infty} m_s(\lambda) = s.$$

Recall from Eq (11) that $0 < b(s) < s$, so (by the intermediate value theorem) there exists a unique finite $\lambda(s)$ satisfying

$$m_s(\lambda(s)) = b(s).$$

Since $m_s(\lambda)$ is measurable in s for every fixed λ and continuous and strictly increasing in λ for ν-almost every s, the unique root $\lambda(s)$ admits a measurable version.

The final assertion of the lemma was established at the start of the proof, where $Z_s(\lambda)$ was shown to be positive and finite for every finite λ. ∎

B The entropy projection onto the S-marginal and Rogers constraints.

The following lemma describes what happens when the (λ, v)-update is applied to a probability density q: the resulting density q' satisfies two of the three constraints, namely the S-marginal and the local Rogers conditions, and for any other density p satisfying these two constraints the relative entropies of p and q' with respect to q are related by the Pythagorean identity $H(p|q) = H(p|q') + H(q'|q)$. In particular q' is the density closest to q in relative entropy among those satisfying the two constraints.

Lemma B.1 *Let $D = \{(s,x) : s \ge X_0,\ 0 < x \le s\}$, and let q be a probability density on D whose S-marginal q^S is positive ν-almost everywhere. For a given measurable function λ which is finite ν-almost everywhere, define*

$$q_\lambda(x|s) = \frac{e^{\lambda(s)x}q(s,x)}{\int_0^s e^{\lambda(s)y}q(s,y)dy}, \qquad x \in (0,s], \text{ for every } s \text{ with } 0 < q^S(s) < \infty,$$

and suppose that λ is such that

$$\int_0^s x q_\lambda(x|s)\,dx = b(s) \qquad \text{for } \nu\text{-almost every } s.$$

Set $q'(s,x) = \nu(s)q_\lambda(x|s)$ for s with $q^S(s) > 0$ and $q'(s,x) = 0$ otherwise, i.e. q' has S-marginal ν and conditional density $q_\lambda(\cdot|s)$ given $S = s$.

Then q' is a probability density on D, and for any probability density p on D with S-marginal ν, conditional mean $\int_0^s xp(x|s)dx = b(s)$ for ν-almost every s, and $H(p|q) < \infty$, we have $H(p|q) = H(p|q') + H(q'|q)$.

Proof. Step 1: properties of q'. For ν-almost every s, we have $q^S(s) > 0$, and integrating the definition of q' over x gives $\int_0^s q'(s,x)dx = \nu(s)$. Hence q' has S-marginal ν and is a probability density on D. Its conditional density given $S = s$ is $q'(x|s) = e^{\lambda(s)x}q(s,x)/\int_0^s e^{\lambda(s)y}q(s,y)dy$, so the defining equation for λ says precisely that $\int_0^s xq'(x|s)dx = b(s)$ for ν-almost every s: q' has the same conditional mean as p. Moreover, dividing by $q(x|s) = q(s,x)/q^S(s)$,

$$\frac{q'(x|s)}{q(x|s)} = \frac{q^S(s)\,e^{\lambda(s)x}}{\int_0^s e^{\lambda(s)y}q(s,y)dy} \qquad \text{for } \nu\text{-almost every } s, \tag{B-1}$$

whose right-hand side is, for ν-almost every s, positive and finite for every $x \in (0,s)$, since $q^S(s) > 0$, $\lambda(s)$ is finite and $q(s,\cdot)$ is integrable. Hence $q'(\cdot|s)$ and $q(\cdot|s)$ vanish on the same set, i.e. they are mutually absolutely continuous, which we write $q(\cdot|s) \sim q'(\cdot|s)$. Taking logarithms, and noting that neither $q^S(s)$ nor the integral in the denominator depends on x,

$$\log\frac{q'(x|s)}{q(x|s)} = \lambda(s)x + c(s), \qquad c(s) := \log q^S(s) - \log\int_0^s e^{\lambda(s)y}q(s,y)dy, \tag{B-2}$$

so the log-density ratio is an affine function of x.

Step 2: decomposing relative entropy along S. Let P and Q be probability densities on D with $P \ll Q$, and write $P(s,x) = P^S(s)P(x|s)$ and $Q(s,x) = Q^S(s)Q(x|s)$. Then $\log \frac{P(s,x)}{Q(s,x)} = \log \frac{P^S(s)}{Q^S(s)} + \log \frac{P(x|s)}{Q(x|s)}$ for P-almost every (s,x), and integrating against P gives

$$H(P|Q) = H(P^S|Q^S) + \int P^S(s) H(P(\cdot|s)|Q(\cdot|s))ds. \tag{B-3}$$

Step 3: the identity fibrewise. Applying Eq (B-3) with $P = p$, $Q = q$ and using $H(p|q) < \infty$, both terms on the right-hand side are finite; in particular $H(p(\cdot|s)|q(\cdot|s)) < \infty$ for ν-almost every s. Fix such an s. Since $q'(\cdot|s)$ is equivalent to $q(\cdot|s)$ by Step 1, Lemma 1.4(b) in [Nutz22] gives

$$H(p(\cdot|s)|q(\cdot|s)) = H(p(\cdot|s)|q'(\cdot|s)) + \int_0^s \log \frac{q'(x|s)}{q(x|s)}\, p(x|s)dx.$$

By Eq (B-2), for any probability density ρ on $(0,s)$ the integral $\int_0^s \log \frac{q'(x|s)}{q(x|s)}\rho(x)dx = \lambda(s)\int_0^s x\rho(x)dx + c(s)$ depends on ρ only through its mean. Since $p(\cdot|s)$ and $q'(\cdot|s)$ both have mean $b(s)$ (by hypothesis and by Step 1 respectively), the last integral above is unchanged if $p(x|s)$ is replaced by $q'(x|s)$, and equals $H(q'(\cdot|s)|q(\cdot|s))$. Therefore

$$H(p(\cdot|s)|q(\cdot|s)) = H(p(\cdot|s)|q'(\cdot|s)) + H(q'(\cdot|s)|q(\cdot|s)) \qquad \text{for } \nu\text{-almost every } s. \tag{B-4}$$

Step 4: integrating over s. Apply Eq (B-3) to the three pairs (p,q), (p,q') and (q',q); the absolute continuity required holds since $p \ll q$ by hypothesis, $q' \ll q$ by Step 1, and $p \ll q'$ because $p(\cdot|s) \ll q(\cdot|s)$ (from $p \ll q$) and $q(\cdot|s) \sim q'(\cdot|s)$ (Step 1, after Eq (B-1)) for ν-almost every s, and both p and q' have S-marginal ν.

Multiplying Eq (B-4) by $\nu(s)$ and integrating over s gives

$$\int \nu(s) H(p(\cdot|s)|q(\cdot|s))ds = \int \nu(s) H(p(\cdot|s)|q'(\cdot|s))ds + \int \nu(s) H(q'(\cdot|s)|q(\cdot|s))ds. \tag{B-5}$$

We now express each of the three integrals in Eq (B-5) via Eq (B-3), whose first term on the right is the relative entropy of the S-marginals. Here $p^S = \nu$ by hypothesis and $q'^S = \nu$ by Step 1, while q^S is the S-marginal of q. Recall (B-3) states that $H(P|Q) = H(P^S|Q^S) + \int P^S(s)H(P(\cdot|s)|Q(\cdot|s))ds$. Then taking $(P,Q) = (p,q)$, (p,q') and (q',q) respectively gives

$$H(p|q) = H(\nu|q^S) + \int \nu(s) H(p(\cdot|s)|q(\cdot|s))ds, \tag{B-6}$$

$$H(p|q') = \int \nu(s) H(p(\cdot|s)|q'(\cdot|s))ds, \tag{B-7}$$

$$H(q'|q) = H(\nu|q^S) + \int \nu(s) H(q'(\cdot|s)|q(\cdot|s))ds. \tag{B-8}$$

Substituting Eqs (B-6), (B-7) and (B-8) into Eq (B-5), i.e. replacing the three integrals there by $H(p|q) - H(\nu|q^S)$, $H(p|q')$ and $H(q'|q) - H(\nu|q^S)$ respectively, the two terms $H(\nu|q^S)$ cancel and we obtain $H(p|q) = H(p|q') + H(q'|q)$. ■

C Well-definedness of the Sinkhorn scheme

For $n \geq 1$, we regard u_n and v_n as extended-real-valued functions, with $u_n(x) = -\infty$ for $x \notin (0,K)$ and $v_n(s) = -\infty$ for $s \notin (X_0, K)$. The multiplier λ_n is only relevant ν-almost everywhere, and we set $\lambda_n(s) = 0$ whenever $\nu(s) = 0$. Thus $p_{n,n,n}(s,x) = R(s,x)e^{u_n(x)+v_n(s)+\lambda_n(s)x}$ vanishes whenever $x \notin (0,K)$ or $s \notin (X_0,K)$.

Lemma C.1 *Suppose Assumption 2.4 holds and set $u_0 = v_0 = \lambda_0 = 0$. Then every step of the Sinkhorn scheme is well defined. For every $n \geq 0$, $H(p^*|p_{n,n,n}) < \infty$, $H(p^*|p_{n+1,n,n}) < \infty$ and*

$$H(p^*|p_{n,n,n}) - H(p^*|p_{n+1,n,n}) = H(p_{n+1,n,n}|p_{n,n,n}), \tag{C-1}$$

$$H(p^*|p_{n+1,n,n}) - H(p^*|p_{n+1,n+1,n+1}) = H(p_{n+1,n+1,n+1}|p_{n+1,n,n}). \tag{C-2}$$

The functions u_n are finite Lebesgue almost everywhere on $(0,K)$, and λ_n and v_n are finite ν-almost everywhere. Every completed iterate $p_{n,n,n}$ with $n \geq 1$ has S-marginal ν and satisfies the local Rogers condition.

Proof. We proceed by induction. Since $p_{0,0,0} = R$, Assumption 2.4 gives $H(p^*|p_{0,0,0}) < \infty$. Also $u_0 = 0$, $\lambda_0 = 0$ and $v_0 = 0$ are finite, so the finiteness assertions hold at $n = 0$. As the inductive hypothesis, we assume that $H(p^*|p_{n,n,n}) < \infty$, u_n is finite Lebesgue almost everywhere on $(0,K)$, and that λ_n and v_n are finite ν-almost everywhere.

Write $q_n = p_{n,n,n}$ and let q_n^X denote its X-marginal. Since the X-marginals of p^* and q_n are μ and q_n^X, respectively, the DPI gives

$$H(\mu|q_n^X) \leq H(p^*|q_n) < \infty. \tag{C-3}$$

Since $H(\mu|q_n^X) < \infty$, we have $\mu \ll q_n^X$, so the set $\{x \in (0,K) : q_n^X(x) = 0\}$ is μ-null; as $\mu > 0$ Lebesgue almost everywhere on $(0,K)$ by Assumption 2.4, this set is Lebesgue-null, i.e. $q_n^X(x) > 0$ for Lebesgue almost every $x \in (0,K)$. Moreover, since q_n^X is a probability density, $q_n^X(x) < \infty$ for Lebesgue almost every $x \in (0,K)$.

Recall the u-update, Eq (8): $u_{n+1}(x) = \log \mu(x) - \log(\int_x^\infty R(s,x)e^{v_n(s)+\lambda_n(s)x}ds)$. Integrating $q_n(s,x) = R(s,x)e^{u_n(x)+v_n(s)+\lambda_n(s)x}$ over s gives $q_n^X(x) = e^{u_n(x)}\int_x^\infty R(s,x)e^{v_n(s)+\lambda_n(s)x}ds$, so the integral inside the logarithm above equals $e^{-u_n(x)}q_n^X(x)$, and therefore

$$u_{n+1}(x) = \log \mu(x) - \log q_n^X(x) + u_n(x)$$

for Lebesgue almost every $x \in (0,K)$. Hence $p_{n+1,n,n}(s,x)/q_n(s,x) = e^{u_{n+1}(x)-u_n(x)} = \mu(x)/q_n^X(x)$, so $p_{n+1,n,n} \ll q_n$, and $p^* \ll q_n$ since $H(p^*|q_n) < \infty$ by the inductive hypothesis.

Lemma 1.4(b) in [Nutz22] gives

$$H(p^*|q_n) - H(p^*|p_{n+1,n,n}) = \mathbb{E}^{p^*}(\log \frac{p_{n+1,n,n}(S,X)}{q_n(S,X)}) = \mathbb{E}^{p^*}(\log \frac{\mu(X)}{q_n^X(X)}).$$

Since the X-marginal of p^* is μ, $\mathbb{E}^{p^*}(\log \frac{\mu(X)}{q_n^X(X)}) = H(\mu|q_n^X)$, and therefore $H(p^*|q_n) - H(p^*|p_{n+1,n,n}) = H(\mu|q_n^X)$. Since $H(p^*|q_n) < \infty$ and $H(\mu|q_n^X) < \infty$, it follows that $H(p^*|p_{n+1,n,n}) < \infty$.

By Eqs (15) and (C-3), $H(p_{n+1,n,n}|q_n) = H(\mu|q_n^X) < \infty$. Hence

$$H(p^*|q_n) - H(p^*|p_{n+1,n,n}) = H(p_{n+1,n,n}|q_n),$$

which is Eq (C-1).

Since $\mu(x) > 0$ and $0 < q_n^X(x) < \infty$ for Lebesgue almost every $x \in (0,K)$ (shown above), and u_n is finite Lebesgue almost everywhere on $(0,K)$ by the inductive hypothesis, the identity $u_{n+1}(x) = \log \mu(x) - \log q_n^X(x) + u_n(x)$ shows that u_{n+1} is finite Lebesgue almost everywhere on $(0,K)$.

We now apply Lemma 2.5 directly to u_{n+1}. Since $p_{n+1,n,n}(s,x) = R(s,x)e^{u_{n+1}(x)+v_n(s)+\lambda_n(s)x}$, its S-marginal is

$$(p_{n+1,n,n})^S(s) = e^{v_n(s)}\int_0^s R(s,x)e^{u_{n+1}(x)+\lambda_n(s)x}dx.$$

Since $p_{n+1,n,n}$ is a probability density, $(p_{n+1,n,n})^S$ is integrable, and hence finite for Lebesgue almost every s. As shown above, $H(p^*|p_{n+1,n,n}) < \infty$. Since the S-marginal of p^* is ν, the DPI gives $H(\nu|(p_{n+1,n,n})^S) \le H(p^*|p_{n+1,n,n}) < \infty$, hence $\nu \ll (p_{n+1,n,n})^S$, i.e. $(p_{n+1,n,n})^S(s) > 0$ for ν-almost every s.

We have already shown that $0 < (p_{n+1,n,n})^S(s) < \infty$ for ν-almost every s. Moreover, v_n is finite ν-almost everywhere by the inductive hypothesis, and hence $0 < e^{-v_n(s)} < \infty$ for ν-almost every s. Therefore

$$0 < \int_0^s R(s,x)e^{u_{n+1}(x)+\lambda_n(s)x}dx < \infty$$

for ν-almost every s. Hence the hypotheses of Lemma 2.5 hold, so the λ-update is well defined and there exists a (ν a.e.) unique finite measurable function $\lambda_{n+1}(s)$ satisfying

$$\frac{\int_0^s xR(s,x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x}dx}{\int_0^s R(s,x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x}dx} = b(s).$$

By Assumption 2.4 and the fact that ν is a probability density, $0 < \nu(s) < \infty$ for ν-almost every s. Now recall Eq (10):

$$v_{n+1}(s) = \log \nu(s) - \log(\int_0^s R(s,x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x}dx).$$

Then $\log \nu(s)$ is finite for ν-almost every s, and the second logarithm in Eq (10) is finite for ν-almost every s by Lemma 2.5 applied with $u = u_{n+1}$ and $\lambda = \lambda_{n+1}(s)$, so the v-update is well defined and v_{n+1} is finite ν-almost everywhere.

We now apply Lemma B.1 with $q = p_{n+1,n,n}$ and $\lambda(s) = \lambda_{n+1}(s) - \lambda_n(s)$. Its hypotheses hold. First, the S-marginal of q is positive ν-almost everywhere (shown above). Second, by definition $q_\lambda(x|s) \propto e^{\lambda(s)x}q(s,x)$; with $\lambda = \lambda_{n+1} - \lambda_n$ and $q = p_{n+1,n,n} = R\,e^{u_{n+1}(x)+v_n(s)+\lambda_n(s)x}$ this is

$$e^{(\lambda_{n+1}(s)-\lambda_n(s))x}\,R(s,x)e^{u_{n+1}(x)+v_n(s)+\lambda_n(s)x} = e^{v_n(s)}\,R(s,x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x},$$

Hence $q_\lambda(x|s) = R(s,x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x}/\int_0^s R(s,y)e^{u_{n+1}(y)+\lambda_{n+1}(s)y}dy$, whose mean $\int_0^s xq_\lambda(x|s)dx$ is the left-hand side of Eq (9), hence equal to $b(s)$. Finally, by the definition of q' in Lemma B.1,

$$\begin{aligned} q'(s,x) = \nu(s)\,q_\lambda(x|s) = \frac{\nu(s)}{\int_0^s R(s,y)e^{u_{n+1}(y)+\lambda_{n+1}(s)y}dy}\,R(s,x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} &= e^{v_{n+1}(s)}R(s,x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} \\ &= p_{n+1,n+1,n+1}(s,x), \end{aligned}$$

where the third equality is Eq (10).

p^* has S-marginal ν, conditional mean b and $H(p^*|p_{n+1,n,n}) < \infty$ (shown above), so Lemma B.1 gives

$$H(p^*|p_{n+1,n,n}) = H(p^*|p_{n+1,n+1,n+1}) + H(p_{n+1,n+1,n+1}|p_{n+1,n,n}) \tag{C-4}$$

which is Eq (C-2). Since the left-hand side is finite and both terms on the right are nonnegative,

$$H(p^*|p_{n+1,n+1,n+1}) < \infty.$$

Finally, $(p_{n+1,n+1,n+1})^S = \nu$ by Lemma B.1, since $p_{n+1,n+1,n+1}$ is the density q' there, and by Eq (9) $\int_0^s x p_{n+1,n+1,n+1}(s,x)dx = b(s)\nu(s) = r(s)$, so the completed iterate satisfies the local Rogers condition. This completes the induction. ■

D Sketch proof of the Rogers' construction

For completeness we sketch the construction in Section 3 of [Rog93] of a continuous martingale with a prescribed joint law of (S_T, X_T), in the case relevant here where the joint law has a density. It is convenient to work with $Y = S - X$ and to write $\bar{\mu}(b,y)$ for the joint density of (S_T, Y_T) on $[x_0,\infty)\times(0,\infty)$, so that $\bar{\mu}(b,y) = \pi(b, b-y)$ in the notation of Theorem 1.1.

Let W be a Brownian motion started at x_0 with running maximum $\bar{W}_t = \max_{u\le t} W_u$, and let $\{Z_b\}_{b\ge x_0}$ be a family of random variables, independent of W and of each other, with Z_b having density $f(\cdot\,;b)$ on $(0,\infty)$. Define $\tau = \inf\{t \ge 0 :\ \bar{W}_t - W_t \ge Z_{\bar{W}_t}\}$. The rate of excursions with drawdown $\ge y$ is $\frac{1}{y}$, so the excursions that reach Z_b arrive at rate $\int_0^\infty y^{-1} f(y;b)dy$. Then

$$\begin{aligned} \mathbb{P}(\bar{W}_\tau > b) &= \exp[-\int_{x_0}^b \int_0^\infty \frac{1}{y} f(y;s)dyds], \\ \mathbb{P}(\bar{W}_\tau \in db,\ \bar{W}_\tau - W_\tau \in dy) &= \mathbb{P}(\bar{W}_\tau \ge b)\,\frac{1}{y} f(y;b)dydb. \end{aligned} \tag{D-1}$$

Now let $\bar{\mu}(b,y)$ be the target density and $\bar{\rho}(b) = \int_b^\infty \int_0^\infty \bar{\mu}(s,u)\,du\,ds = \mathbb{P}(S_T > b)$. Matching (D-1) with $\bar{\mu}$ requires $\mathbb{P}(\bar{W}_\tau \ge b) = \bar{\rho}(b)$ and

$$f(y;b) = \frac{y\bar{\mu}(b,y)}{\bar{\rho}(b)}.$$

This definition is used only for levels b with $\bar{\rho}(b) > 0$; at other levels the mark distribution may be chosen arbitrarily.

Since $\nu(b) = \int_0^\infty \bar{\mu}(b,y)dy$, we have $\int_0^\infty y^{-1} f(y;b)dy = \nu(b)/\bar{\rho}(b) = -\bar{\rho}'(b)/\bar{\rho}(b)$. Hence, replacing the inner integral in the first identity above and using $\bar{\rho}(x_0) = 1$ gives

$$\mathbb{P}(\bar{W}_\tau > b) = \exp(\int_{x_0}^b \frac{\bar{\rho}'(s)}{\bar{\rho}(s)}ds) = \frac{\bar{\rho}(b)}{\bar{\rho}(x_0)} = \bar{\rho}(b). \tag{D-2}$$

This is a legitimate choice only if $f(\cdot\,;b)$ is a probability density in y for each b, i.e.

$$\int_0^\infty y\bar{\mu}(b,y)dy = \bar{\rho}(b) \qquad \text{for almost every } b \ge x_0. \tag{D-3}$$

Substituting $\bar{\mu}(b,y) = \pi(b,b-y)$, the left-hand side of (D-3) is $\int_{-\infty}^b (b-x)\pi(b,x)dx = b\nu(b) - \eta(b)$ and $\bar{\rho}(b) = \bar{F}_\nu(b)$, so (D-3) is exactly the local Rogers condition (1). $\bar{\rho}(b) \to 0$ as $b \to \infty$, the stopping time τ is almost surely finite.

The constructed joint law is the prescribed law π. Its integrability, mean and tail conditions in Theorem 1.1 therefore imply that $(W_{t\wedge\tau})_{t\ge0}$ is uniformly integrable, by Lemma 2.3 of [Rog93], applied after subtracting x_0.

E Proof of

To see why $b(.)$ is strictly increasing, recall that $\bar{F}'_\nu = -\nu$. First, $\frac{d}{ds}\frac{1}{\bar{F}_\nu(s)} = -\frac{\bar{F}'_\nu(s)}{\bar{F}_\nu(s)^2} = \frac{\nu(s)}{\bar{F}_\nu(s)^2}$, so $1/\bar{F}_\nu$ is strictly convex if and only if $s \mapsto \nu(s)/\bar{F}_\nu(s)^2$ is strictly increasing, in which case (taking logs) its derivative is non-negative and vanishes on no interval:

$$\frac{d}{ds}\log\frac{\nu(s)}{\bar{F}_\nu(s)^2} = \frac{\nu'(s)}{\nu(s)} - 2\frac{\bar{F}'_\nu(s)}{\bar{F}_\nu(s)} = \frac{\nu'(s)}{\nu(s)} + \frac{2\nu(s)}{\bar{F}_\nu(s)} \ge 0.$$

Second, by the quotient rule,

$$b'(s) = 1 - \frac{\bar{F}'_\nu(s)\nu(s) - \bar{F}_\nu(s)\nu'(s)}{\nu(s)^2} = 1 + \frac{\nu(s)^2 + \bar{F}_\nu(s)\nu'(s)}{\nu(s)^2} = 2 + \frac{\bar{F}_\nu(s)\,\nu'(s)}{\nu(s)^2}.$$

Multiplying by $\nu(s)/\bar{F}_\nu(s) > 0$, $b'(s)$ has the sign of $\frac{2\nu(s)}{\bar{F}_\nu(s)} + \frac{\nu'(s)}{\nu(s)}$, the quantity just obtained. Hence $b' \geq 0$, with $b' > 0$ outside a set with empty interior, so b is strictly increasing.

We break the remainder of the proof into two parts:

(i) Since b is strictly increasing on the support of ν, S_T and $Z = b(S_T)$ generate the same σ-algebra, so the local Rogers condition (1) can be rewritten as $\mathbb{E}(X_T|S_T) = b(S_T) \iff \mathbb{E}(X_T|Z) = Z$, i.e. a joint law of (S_T, X_T) with marginals ν and μ satisfies the local Rogers condition if and only if the law of (Z, X_T) is a martingale coupling of ν_Z and μ. By Strassen's theorem, such a martingale coupling m exists if and only if $\mu \succeq_{\mathrm{cx}} \nu_Z$.

Given such an m, let π be the law of (S_T, X_T) when $(Z, X_T) \sim m$ and $S_T = b^{-1}(Z)$, i.e. $\pi(ds, dx) = \nu(ds)\, m_{b(s)}(dx)$. By the equivalence above, π has marginals ν and μ and satisfies the local Rogers condition, and the remaining conditions of Theorem 1.1 (which concern only the marginals) hold by Assumption 2.1. Hence π is admissible if and only if it also satisfies the support condition $x \leq s$, i.e. if and only if $m_{b(s)}$ charges only $(-\infty, s]$ for ν-almost every s, which is the stated condition on m_z with $z = b(s)$.

(ii) By [BJ16], under $\mu \succeq_{\mathrm{cx}} \nu_Z$ the left-curtain coupling of (ν_Z, μ) exists and by [HLT16] it is the two-point kernel $m_z = q\, \delta_{T_u(z)} + (1-q)\, \delta_{T_d(z)}$, $q = (z - T_d(z))/(T_u(z) - T_d(z))$, whose branches $T_d(z) \leq z \leq T_u(z)$ are characterised by the two identities

$$\mu([T_d(z), T_u(z)]) \;=\; \nu_Z((b(x_0), z]), \qquad \int_{T_d(z)}^{T_u(z)} x\mu(dx) \;=\; \int_{b(x_0)}^{z} z'\nu_Z(dz').$$

Substituting $z = b(s)$, $x_\pm(s) = T_{u/d}(b(s))$, and using $\nu_Z((b(x_0), b(s)]) = \nu((x_0, s]) = 1 - \bar{F}_\nu(s)$ and $\int_{b(x_0)}^{b(s)} z'\nu_Z(dz') = \int_{x_0}^{s} b(u)\nu(u)du$, these are the two displayed equations in the statement.